6. Energy-based Model

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FS26 MTH345: Deep Generative Models: A Statistical Perspective

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Lab Homepage

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

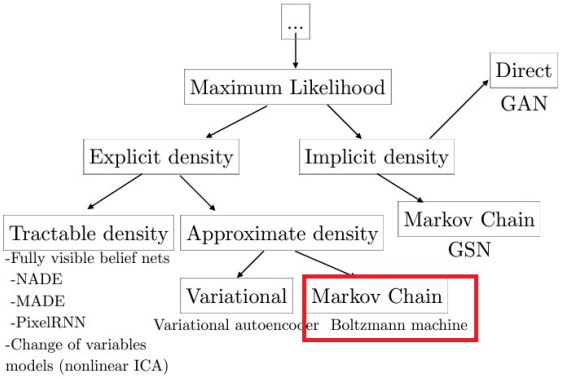
M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- Autoregressive models can flexibly model complex joint structures, but their sequential generation process is time-consuming.
- In this lecture, we will review *energy-based models*, another class of latent variable-based generative models.
- Under certain conditions, these models can universally approximate arbitrary distributions and offer faster data generation than autoregressive models.

Lecture Topics



- Recommended reference materials include:
  - Hinton (2002), "Training products of experts by minimizing contrastive divergence," *Neural computation*.
  - Goodfellow (2016), "Nips 2016 tutorial: Generative adversarial networks," *arXiv*.

The figure is edited from Goodfellow (2016).

Limitations of Bayesian Networks



The images are from the Street View House Numbers (SVHN) dataset (Netzer et al., 2011).

Limitations of Bayesian Networks



- Let us assume, as part of the historical background, that modern houses have progressively transitioned to using blue-backgrounded signs.
- Let Z_1 indicate whether the house has an administrative origin (e.g., conforming to modern building codes), and Z_2 indicate whether the digit is the ending one.
- Given $\boldsymbol{X}$, Z_1 and Z_2 naturally preserve (conditional) independence, i.e.,

$$Z_1 \perp\!\!\!\perp Z_2 \mid \boldsymbol{X}. \tag{1}$$

Limitations of Bayesian Networks

- Many latent variable models, including ICA, are formulated as Bayesian networks that satisfy

$$f(\boldsymbol{z}) = \prod_{j=1}^{r} f(z_j) \quad \text{and} \quad f_\theta(\boldsymbol{x} \mid \boldsymbol{z}) = \prod_{j=1}^{d} f_\theta(x_j \mid \boldsymbol{z}). \tag{2}$$

- Although the prior latent components $\boldsymbol{z}$ are statistically independent, after conditioning on the observed data $\boldsymbol{x}$, the posterior ones can become dependent:

$$f_\theta(\boldsymbol{z} \mid \boldsymbol{x}) = \frac{\prod_{j=1}^{d} f_\theta(x_j \mid \boldsymbol{z}) \prod_{j=1}^{r} f(z_j)}{f_\theta(\boldsymbol{x})} \neq \prod_{j=1}^{r} f_\theta(z_j \mid \boldsymbol{x}) \tag{3}$$

This phenomenon is known as *explaining away* in directed graphical models.

Energy-based Models

- Restricted Boltzmann machines (RBMs) are latent variable models that guarantee conditional independence among variables within the same layer.
- We first review energy-based models and general Boltzmann machines.
- *Energy-based models* are generative models that utilize an energy function to define a *Boltzmann distribution*[1]:

$$f_\theta(\boldsymbol{x}) := \frac{\exp(-E_\theta(\boldsymbol{x}))}{C(\theta)} \tag{4}$$

where $E_\theta(\boldsymbol{x})$ denotes the energy function, and $C(\theta) := \int \exp(-E_\theta(\boldsymbol{x}))d\boldsymbol{x}$ is the partition function (or normalization constant).

[1] Here, we assume unit temperature and a unit Boltzmann constant.

Energy-based Models

- For example, the multivariate Gaussian distribution with parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is given by

$$f_\theta(\boldsymbol{x}) = (2\pi)^{-d/2}|\boldsymbol{\Sigma}|^{-1/2}\exp\big(-(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})/2\big), \tag{5}$$

  which yields the energy function

$$E_\theta(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}). \tag{6}$$

- Its corresponding partition function (or normalization constant) is $C(\theta) = (2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}$.[2]

[2] In general, any distribution in the exponential family, represented by $p_\theta(\boldsymbol{x}) = h(\boldsymbol{x})\exp\big(\boldsymbol{\eta}(\theta)^T\boldsymbol{T}(\boldsymbol{x}) - A(\theta)\big)$, can be interpreted as an instance of Equation (4). In this case, the energy function is $E_\theta(\boldsymbol{x}) = -\boldsymbol{\eta}(\theta)^T\boldsymbol{T}(\boldsymbol{x}) - \log h(\boldsymbol{x})$ and the partition function is $C(\theta) = \exp(A(\theta))$.

Boltzmann Machine

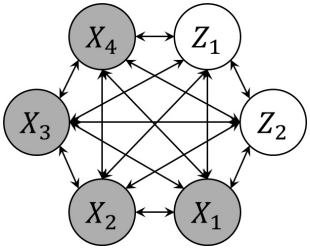
- Boltzmann machines (Ackley et al., 1985) are classical energy-based models that introduce latent variables to define a joint energy density function over both the latent variables and observations.
- The energy function of a Boltzmann machine is expressed as

$$E_\theta(\boldsymbol{z}, \boldsymbol{x}) := -\left(\frac{1}{2} \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix}^T \boldsymbol{W} \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^T \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix} \right), \tag{7}$$

  where $\theta := (\boldsymbol{W}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ represents the parameters. Here, $\boldsymbol{W}$ is a symmetric matrix with zero diagonal elements, indicating the absence of self-loops.
- The corresponding joint density is then given by

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) \propto \exp\left(\frac{1}{2} \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix}^T \begin{pmatrix} \boldsymbol{W}_{ZZ} & \boldsymbol{W}_{ZX} \\ \boldsymbol{W}_{ZX}^T & \boldsymbol{W}_{XX} \end{pmatrix} \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^T \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix} \right). \tag{8}$$

Boltzmann Machine



- For example, when we have $\boldsymbol{x} \in \{0,1\}^{d=4}$ and $\boldsymbol{z} \in \{0,1\}^{r=2}$, the joint density function can be explicitly expanded as:

$$p_\theta(\boldsymbol{z}, \boldsymbol{x})$$

$$\propto \exp\left(W_{ZZ,1,2} z_1 z_2 + \sum_{i=1}^{2} \sum_{j=1}^{4} W_{ZX,i,j} z_i x_j + \sum_{j<k} W_{XX,j,k} x_j x_k + \sum_{i=1}^{2} \alpha_i z_i + \sum_{j=1}^{4} \beta_j x_j \right). \tag{9}$$

Boltzmann Machine

Exercise 1

Consider a Boltzmann machine with d binary visible units $\boldsymbol{x} \in \{0,1\}^d$ and r binary hidden units $\boldsymbol{z} \in \{0,1\}^r$:

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) = \frac{1}{C(\theta)} \exp\left(-E_\theta(\boldsymbol{z}, \boldsymbol{x})\right), \tag{10}$$

where the energy function is defined as

$$E_\theta(\boldsymbol{z}, \boldsymbol{x}) = -\frac{1}{2} \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix}^\top \begin{pmatrix} \boldsymbol{W}_{ZZ} & \boldsymbol{W}_{ZX} \\ \boldsymbol{W}_{ZX}^\top & \boldsymbol{W}_{XX} \end{pmatrix} \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix} - \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^\top \begin{pmatrix} \boldsymbol{z} \\ \boldsymbol{x} \end{pmatrix}, \tag{11}$$

with symmetric weight matrices having zero diagonals ($\mathrm{diag}(\boldsymbol{W}_{ZZ}) = 0$ and $\mathrm{diag}(\boldsymbol{W}_{XX}) = 0$).

Boltzmann Machine

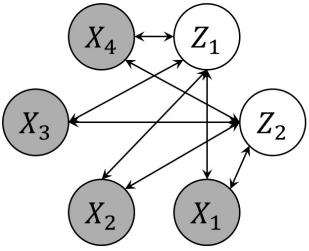
For a given $\boldsymbol{x}$, define $\boldsymbol{x}_{j,\text{on}} := (x_1, \ldots, x_j = 1, \ldots, x_d)^\top$ and $\boldsymbol{x}_{j,\text{off}} := (x_1, \ldots, x_j = 0, \ldots, x_d)^\top$. Show that:

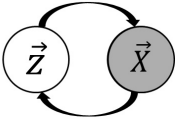
$$\log\left(\frac{p_\theta(\boldsymbol{z}, \boldsymbol{x}_{j,\text{on}})}{p_\theta(\boldsymbol{z}, \boldsymbol{x}_{j,\text{off}})}\right) = \left(\boldsymbol{W}_{ZX}^\top \boldsymbol{z} + \boldsymbol{W}_{XX}\boldsymbol{x}_{j,\text{off}} + \boldsymbol{\beta}\right)_j. \tag{12}$$

Boltzmann Machine



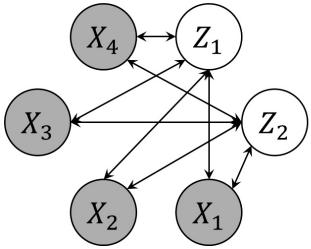
Restricted Boltzmann Machine

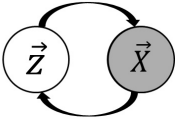




- Restricted Boltzmann machines (RBMs) simplify general Boltzmann machines by eliminating connections within latent variables and within observed features, thereby defining a bipartite graph.
- They were originally proposed by Smolensky et al. (1986) under the name *Harmonium* and gained widespread popularity after Hinton (2002) introduced efficient training algorithms.

Restricted Boltzmann Machine





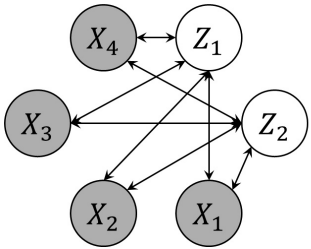
- The energy function of an RBM is explicitly expressed as

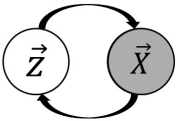
$$E_\theta(\boldsymbol{z}, \boldsymbol{x}) := -\left(\boldsymbol{z}^T \boldsymbol{W} \boldsymbol{x} + \boldsymbol{\alpha}^T \boldsymbol{z} + \boldsymbol{\beta}^T \boldsymbol{x}\right), \tag{13}$$

  which directly corresponds to Equation (8) under the structural constraints $\boldsymbol{W}_{ZZ} = \mathrm{O}$, $\boldsymbol{W}_{XX} = \mathrm{O}$, and $\boldsymbol{W}_{ZX} = \boldsymbol{W}$.[3]

[3] RBMs traditionally assume binary observations and binary latent variables, but they can be extended to generic exponential family distributions (Welling et al., 2004).

Restricted Boltzmann Machine





- For example, when we have $\boldsymbol{x} \in \{0,1\}^{d=4}$ and $\boldsymbol{z} \in \{0,1\}^{r=2}$, the joint density $p_\theta(\boldsymbol{z}, \boldsymbol{x})$ simplifies to:

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) \propto \exp\left(\sum_{i=1}^{2} \sum_{j=1}^{4} W_{i,j} z_i x_j + \sum_{i=1}^{2} \alpha_i z_i + \sum_{j=1}^{4} \beta_j x_j \right). \tag{14}$$

Restricted Boltzmann Machine

Exercise 2

Consider an RBM with d binary visible units $\boldsymbol{x} \in \{0,1\}^d$ and r binary hidden units $\boldsymbol{z} \in \{0,1\}^r$:

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) = \frac{1}{C(\theta)} \exp\left(-E_\theta(\boldsymbol{z}, \boldsymbol{x})\right), \tag{15}$$

where the energy function is defined as $E_\theta(\boldsymbol{z}, \boldsymbol{x}) := -\left(\boldsymbol{z}^T \boldsymbol{W} \boldsymbol{x} + \boldsymbol{\alpha}^T \boldsymbol{z} + \boldsymbol{\beta}^T \boldsymbol{x}\right)$.

Restricted Boltzmann Machine

1. Show that $p_\theta(\boldsymbol{z}, \boldsymbol{x})$ exhibits an exact conditional independence property:

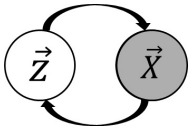
$$p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) = \prod_{j=1}^{d} p_\theta(x_j \mid \boldsymbol{z}) \quad \text{and} \quad p_\theta(\boldsymbol{z} \mid \boldsymbol{x}) = \prod_{i=1}^{r} p_\theta(z_i \mid \boldsymbol{x}). \tag{16}$$

Restricted Boltzmann Machine

2. Show that:

$$\log \left(\frac{p_\theta(z_1, \ldots, z_j = 1, \ldots, z_r \mid \boldsymbol{x})}{p_\theta(z_1, \ldots, z_j = 0, \ldots, z_r \mid \boldsymbol{x})} \right) = (\boldsymbol{W}\boldsymbol{x} + \boldsymbol{\alpha})_j. \tag{17}$$

Restricted Boltzmann Machine



- RBMs generate synthetic data via the following M-step Gibbs sampling process:
  1. Initialize by sampling $\tilde{\boldsymbol{X}}_1$ from a pre-specified distribution $p(\tilde{\boldsymbol{x}})$[4] and draw $\tilde{\boldsymbol{Z}}_1 \sim p_\theta(\boldsymbol{z} \mid \boldsymbol{x} = \tilde{\boldsymbol{X}}_1)$.
  2. For $m = 1, \ldots, M-1$, sequentially generate $\tilde{\boldsymbol{X}}_{m+1} \sim p_\theta(\boldsymbol{x} \mid \boldsymbol{z} = \tilde{\boldsymbol{Z}}_m)$ and $\tilde{\boldsymbol{Z}}_{m+1} \sim p_\theta(\boldsymbol{z} \mid \boldsymbol{x} = \tilde{\boldsymbol{X}}_{m+1})$.
  3. The resulting pair $(\tilde{\boldsymbol{Z}}_M, \tilde{\boldsymbol{X}}_M)$ constitutes the M-step Gibbs sample. Under certain regularity conditions, the joint density of $(\tilde{\boldsymbol{Z}}_M, \tilde{\boldsymbol{X}}_M)$ converges to the true joint distribution $p_\theta(\boldsymbol{z}, \boldsymbol{x})$ as $M \to \infty$.

[4] A common choice is to utilize the empirical distribution of real observations $\boldsymbol{X}$.

Contrastive Divergence

- (Restricted) Boltzmann machines aim to find maximum likelihood estimators. However, the log-likelihood function,

$$\log p_\theta(\boldsymbol{x}) = -\log C(\theta) + \log \sum_{\boldsymbol{z}} \exp\big(-E_\theta(\boldsymbol{z}, \boldsymbol{x})\big), \tag{18}$$

  contains the intractable $C(\theta)$, posing a severe computational challenge.
- Contrastive divergence (CD) (Hinton, 2002) circumvents this issue by evaluating the gradient of the log-likelihood (the score function) directly:

$$\nabla_\theta \log p_\theta(\boldsymbol{x}) = -\underbrace{\sum_{\boldsymbol{z}} \Big(\nabla_\theta E_\theta(\boldsymbol{z}, \boldsymbol{x})\Big) p_\theta(\boldsymbol{z} \mid \boldsymbol{x})}_{\text{Positive gradient}} + \underbrace{\sum_{\boldsymbol{x}, \boldsymbol{z}} \Big(\nabla_\theta E_\theta(\boldsymbol{z}, \boldsymbol{x})\Big) p_\theta(\boldsymbol{z}, \boldsymbol{x})}_{\text{Negative gradient}}. \tag{19}$$

- The score function is exactly the contrast between the positive and negative phases, which is the direct origin of the name *contrastive divergence*.

Contrastive Divergence

Exercise 3

Consider an RBM with d binary visible units $\boldsymbol{x} \in \{0,1\}^d$ and r binary hidden units $\boldsymbol{z} \in \{0,1\}^r$:

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) = \frac{1}{C(\theta)} \exp\left(-E_\theta(\boldsymbol{z}, \boldsymbol{x})\right), \tag{20}$$

where the energy function is defined as $E_\theta(\boldsymbol{z}, \boldsymbol{x}) := -\left(\boldsymbol{z}^T \boldsymbol{W} \boldsymbol{x} + \boldsymbol{\alpha}^T \boldsymbol{z} + \boldsymbol{\beta}^T \boldsymbol{x}\right)$. Show that:

$$\nabla_\theta \log p_\theta(\boldsymbol{x}) = -\underbrace{\sum_{\boldsymbol{z}} \Big(\nabla_\theta E_\theta(\boldsymbol{z}, \boldsymbol{x})\Big) p_\theta(\boldsymbol{z} \mid \boldsymbol{x})}_{\text{Positive gradient}} + \underbrace{\sum_{\boldsymbol{x}, \boldsymbol{z}} \Big(\nabla_\theta E_\theta(\boldsymbol{z}, \boldsymbol{x})\Big) p_\theta(\boldsymbol{z}, \boldsymbol{x})}_{\text{Negative gradient}}. \tag{21}$$

Contrastive Divergence



Contrastive Divergence

- Driven by this formulation, the CD algorithm applies short-run Gibbs sampling to approximate the intractable expectations in both phases.
- For a given training dataset $(\boldsymbol{x}_i)_{i=1}^n$, the execution of the CD algorithm is formalized as follows:
  1. **Positive Phase:** For a given training observation $\boldsymbol{x}_i$, sample $\tilde{\boldsymbol{z}}_i \sim p_\theta(\boldsymbol{z} \mid \boldsymbol{x} = \boldsymbol{x}_i)$ to construct the i-th *positive sample* $(\tilde{\boldsymbol{z}}_i, \boldsymbol{x}_i)$.
  2. **Negative Phase:** Starting from $\boldsymbol{x}_i$, draw the i-th *negative sample* $(\tilde{\boldsymbol{z}}_{i,M}, \tilde{\boldsymbol{x}}_{i,M})$ using M-step Gibbs sampling.
  3. **Gradient Evaluation:** Compute the empirical positive and negative gradients using the sampled mini-batches.
  4. **Parameter Update:** Update θ by ascending the approximated log-likelihood gradient.

Generation Result

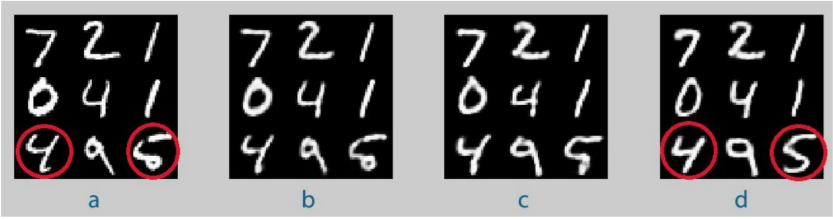


Figure 8: Results from an RBM with 250 hidden units that has been trained on MNIST dataset. In this experiment, after extracting 250 features from 9 MNIST images (28*28 pixel), the new images have been generated from extracted features. (a) 9 MNIST images. (b) Generated images from extracted features with $k = 1$ sampling iteration. (c) Generated images from extracted features with $k = 10$ sampling iterations. (d) Generated images from extracted features with $k = 100$ sampling iterations. The related code is in "*test_ generateData.m*".

Images are from Keyvanrad and Homayounpour (2014).

Flexibility vs. Interpretability

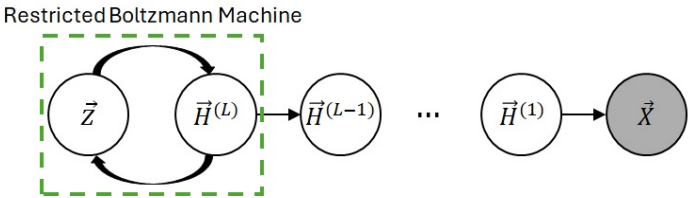
- Compared with general Boltzmann machines, RBMs sacrifice intra-layer flexibility by eliminating connections within $\boldsymbol{z}$ and within $\boldsymbol{x}$.
- Nonetheless, RBMs function as distributional universal approximators:

Theorem (Representational Power of RBMs; Le Roux and Bengio, 2008)

For any $\epsilon > 0$ and any target probability density function $p(\boldsymbol{x})$ supported on $\{0,1\}^d$, there exists an RBM parameterized by θ such that

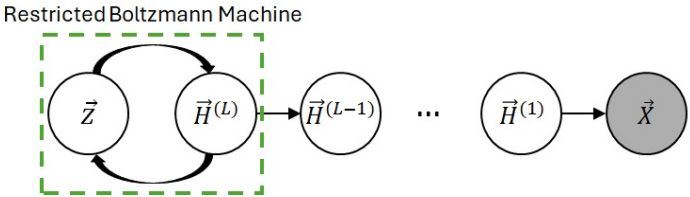
$$\mathcal{D}_{\mathrm{KL}}\big(p(\boldsymbol{x}) \parallel p_\theta(\boldsymbol{x})\big) < \epsilon. \tag{22}$$

Deep Belief Network



- Motivated by RBMs, several of the earliest deep generative models have been proposed, including deep belief networks (DBNs) (Bengio et al., 2006) and deep Boltzmann machines (Salakhutdinov and Hinton, 2009).

Deep Belief Network



- Deep belief networks (Bengio et al., 2006) extend RBMs by stacking hidden layers.

$$p_\theta(\boldsymbol{z}, \boldsymbol{h}^{(L)}, \boldsymbol{h}^{(L-1)}, \ldots, \boldsymbol{h}^{(1)}, \boldsymbol{x}) := p_\theta(\boldsymbol{z}, \boldsymbol{h}^{(L)}) p_\theta(\boldsymbol{h}^{(L-1)}|\boldsymbol{h}^{(L)}) p_\theta(\boldsymbol{h}^{(1)}|\boldsymbol{h}^{(2)}) p_\theta(\boldsymbol{x}|\boldsymbol{h}^{(1)}). \tag{23}$$

Here, $p_\theta(\boldsymbol{h}^{(l-1)}|\boldsymbol{h}^{(l)}) = \prod_i p_\theta(\boldsymbol{h}_i^{(l-1)}|\boldsymbol{h}^{(l)})$ and

$$\boldsymbol{H}_i^{(l-1)}|\boldsymbol{H}^{(l)} \sim \mathrm{Ber}\big(\mathrm{sig}(\boldsymbol{b}^{(l)} + W^{(l)}\boldsymbol{H}^{(l)})\big). \tag{24}$$

Deep Belief Network: Reconstruction Result

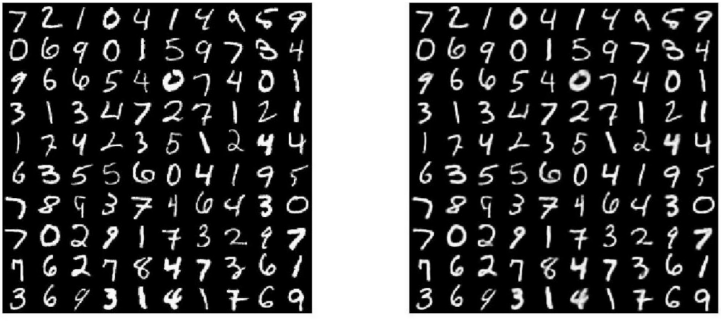
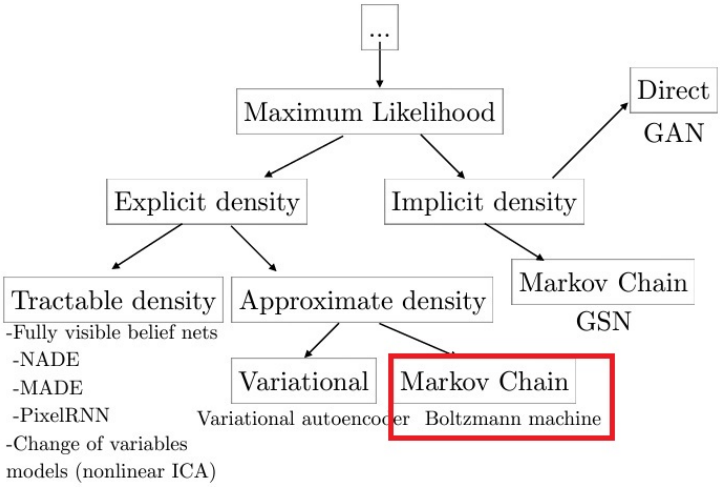


Figure 7: Plotting 100 samples with *plotData* function in *DataStore* class. The first image is 100 samples from MNIST dataset and the second one is reconstructed samples with a DBN model. The related code is in "*test_plotData.m*" file.

Images are from Keyvanrad and Homayounpour (2014).

Lecture Topics



The figure is edited from Goodfellow (2016).

References I

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